Chapter 1 PHYSICAL QUANTITIES AND VECTORS

- 1.1 The Nature of Physics
- 1.2 Idealized Models
- 1.3 Standareds and Units
- 1.4 Uncertainty and Significant Figures
- 1.5 Vetors and Scalars
- 1.6 Vector Algebra
- 1.7 Unit Vectors and Component Expression

Goals for chapter 1

- Three fundamental quantities of physics and the units physicists use to measure them.
- How to keep track of significant figures in your calculations.
- The difference between scalars and vectors, and how to add and subtract vectors graphically.
- What the components of a vector are, and how to use them in calculations.
- What unit vectors are, and how to use them with components to describe vectors.

Introduction

- Being able to predict the path of a thunderstorm is essential for minimizing the damage it does to lives and property. If a thunderstorm is moving at 20 km/h in a direction 53° north of east, how far north does the thunderstorm move in 1 h?

1.1 The Nature of Physics

- Physics is an **experimental** science in which physicists seek patterns that relate the phenomena of nature.
- The patterns are called **physical theories**.
- A very well established or widely used theory is called a **physical law** or **principle**.

According to legend, Galileo investigated falling objects by dropping them from the Leaning Tower of Pisa, Italy, ...

... and he studied pendulum motion by observing the swinging chandelier in the adjacent cathedral.

Solving Problems in Physics

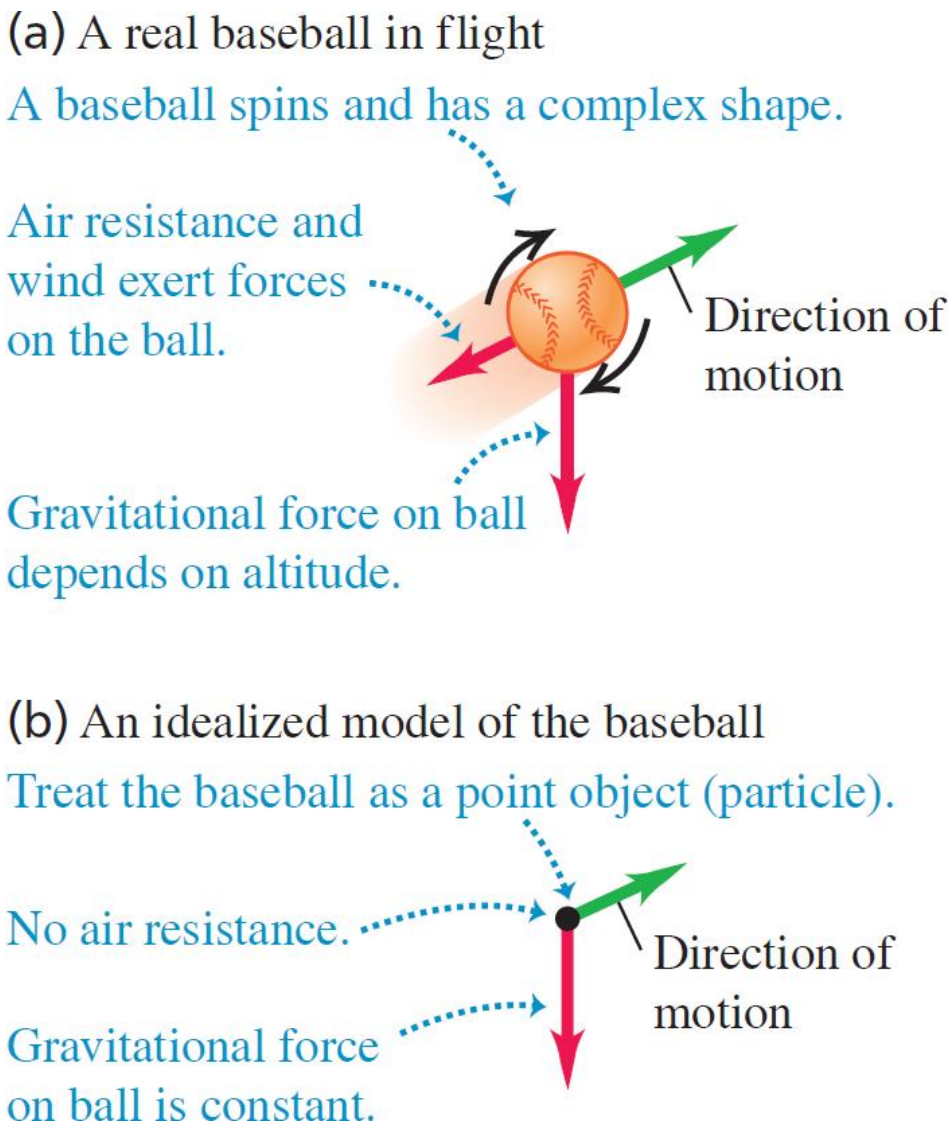
- All of the **Problem-Solving Strategies** and **Examples** in this book will follow these four steps:
- **Identify** the relevant concepts, target variables, and known quantities, as stated or implied in the problem.
- **Set Up** the problem: Choose the equations that you'll use to solve the problem, and draw a sketch of the situation.
- **Execute** the solution: This is where you "do the math."
- **Evaluate** your answer: Compare your answer with your estimates, and reconsider things if there's a discrepancy.

1.2 Idealized Models

To simplify the analysis of

a) a baseball in flight, we use

b) an idealized model.



1.3 Standards and Units

- Length, time, and mass are three **fundamental** quantities of physics.
- The **International System** (SI for **Système International**) is the most widely used system of units.
- In SI units, length is measured in **meters**, time in **seconds**, and mass in **kilograms**.

- **Physical quantity=numerical value + unit e.g., 18.3km**

- **Standard International (SI) units:**

Length (meter):

The meter is the length of path traveled by light in vacuum during a time interval of 1/299,792,458 of a second.

Mass (kilogram):

The standard mass is a particularplatinum–iridium cylinder, kept at the International Bureau of Weights and Measures near Paris, France, whose mass is defined as exactly 1 kg.

Time (second):

The standard second is now defined more precisely in terms of the frequency of radiation emitted by cesium atoms when they pass between two particular states. Specifically, one second is defined as the time required for 9,192,631,770 periods of this radiation.

Unit Prefixes

- Prefixes can be used to create larger and smaller units for the fundamental quantities. Some examples are:

$$Length$$

$$1 nanometer = 1 nm = 10^{-9} m$$ (a few times the size of the largest atom)

$$1 micrometer = 1 \mu m = 10^{-6} m$$ (size of some bacteria and living cells)

$$1 millimeter = 1 mm = 10^{-3} m$$ (diameter of the point of a ballpoint pen)

$$1 centimeter = 1 cm = 10^{-2} m$$ (diameter of your little finger)

$$1 kilometer = 1 km = 10^{3} m$$ (a 10-minute walk)

Unit Prefixes

$Mass$

$1microgram = 1\mu g = 10^{-6} g = 10^{-9} kg$ (mass of a very small dust particle)

$1milligram = 1mg = 10^{-3} g = 10^{-6} kg$ (mass of a grain of salt)

$1gram = 1g = 10^{-3} kg$ (mass of a paper clip)

$Mass$

$1nanosecond = 1ns = 10^{-9} s$ (time for light to travel 0.3 m)

$1microsecond = 1\mu s = 10^{-6} s$ (time for an orbiting space shuttle to travel 8 mm)

$1millisecond = 1ms = 10^{-3} s$ (time for sound to travel 0.35 m)

Unit Consistency and Conversions

- An equation must be **dimensionally consistent**. Terms to be added or equated must **always** have the same units. (Be sure you're adding "apples to apples.")
- Always carry units through calculations.
- Convert to standard units as necessary, by forming a ratio of the same physical quantity in two different units, and using it as a multiplier.
- For example, to find the number of seconds in 3 min, we write:

$$3\ \text{min} = (3\ \cancel{\text{min}})\left(\frac{60\ \text{s}}{1\ \cancel{\text{min}}}\right) = 180\ \text{s}$$

1.4 Uncertainty and Significant Figures

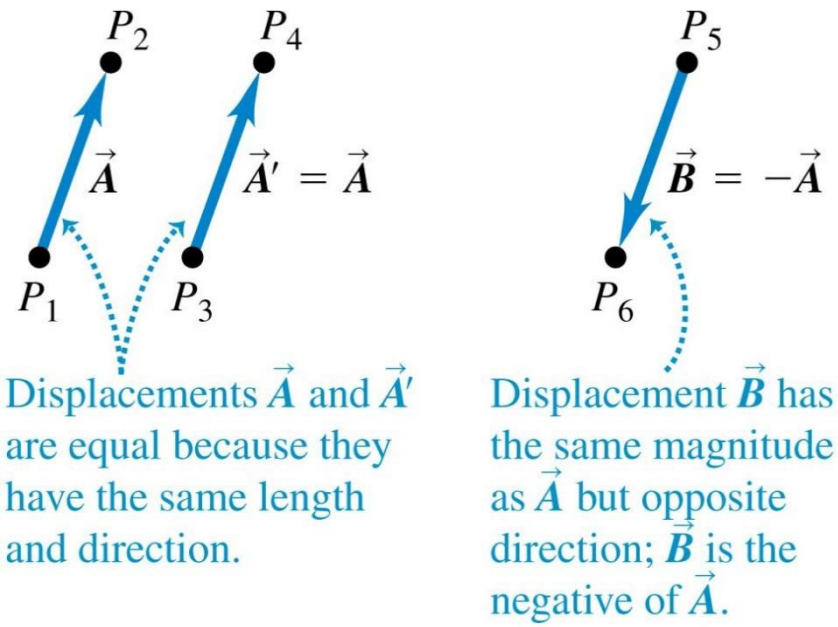
- The uncertainty of a measured quantity is indicated by its number of **significant figures**.
- For multiplication and division, the answer can have no more significant figures than the **smallest** number of significant figures in the factors.
- For addition and subtraction, the number of significant figures is determined by the term having the fewest digits to the right of the decimal point.
- As this train mishap illustrates, even a small percent error can have spectacular results!

1.5 Vectors and Scalars

- A **scalar quantity** can be described by a **single number**.
- A **vector quantity** has both a **magnitude** and a **direction** in space.
- In this book, a vector quantity is represented in boldface italic type with an arrow over it: $\vec{\boldsymbol{A}}$.
- The magnitude of $\vec{\boldsymbol{A}}$ is written as A or $\left|\vec{\boldsymbol{A}}\right|$.

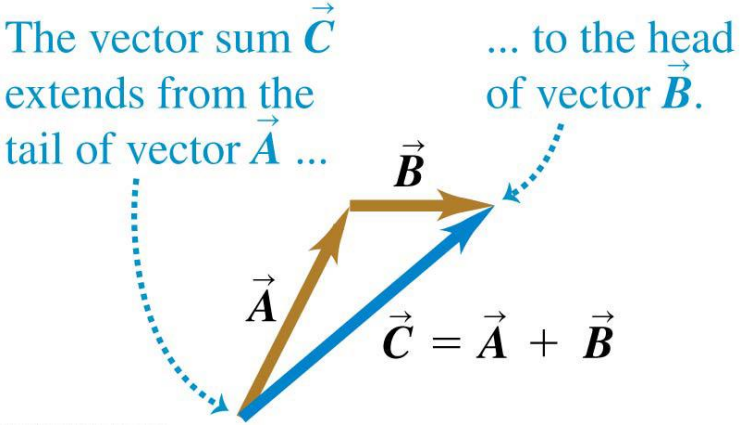
1.6 Vector Algebra

- Draw a vector as a line with an arrowhead at its tip.
- The **length** of the line shows the vector's **magnitude**.
- The **direction** of the line shows the vector's **direction**.

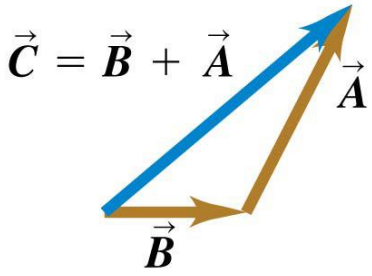


Vector addition

(a) We can add two vectors by placing them head to tail.

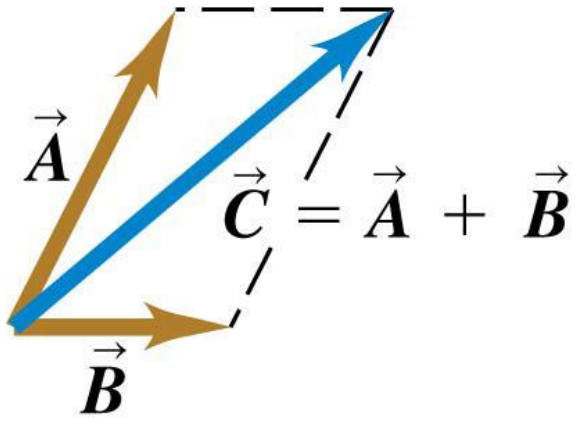


(b) Adding them in reverse order gives the same result: $\vec{A} + \vec{B} = \vec{B} + \vec{A}$. The order doesn't matter in vector addition.



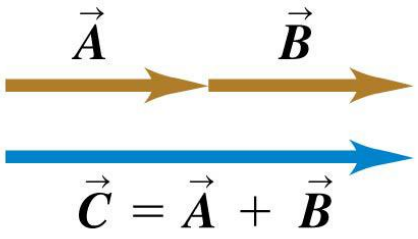
Adding Two Vectors Graphically (2 of 3)

(c) We can also add two vectors by placing them tail to tail and constructing a parallelogram.

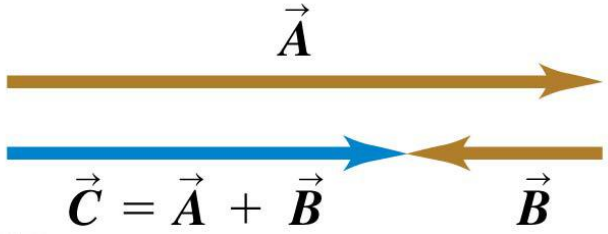


Adding Two Vectors Graphically (3 of 3)

(a) Only when vectors $\vec{A}$ and $\vec{B}$ are parallel does the magnitude of their vector sum $\vec{C}$ equal the sum of their magnitudes: $C = A + B$.



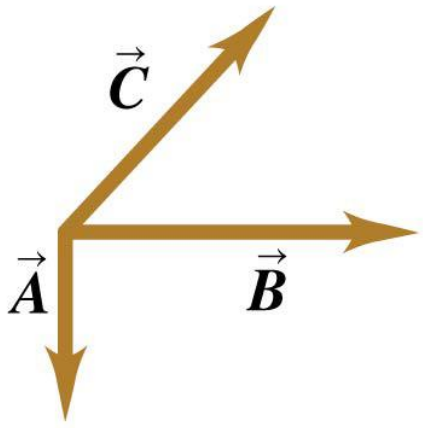
(b) When $\vec{A}$ and $\vec{B}$ are antiparallel, the magnitude of their vector sum $\vec{C}$ equals the *difference* of their magnitudes: $C = |A - B|$.



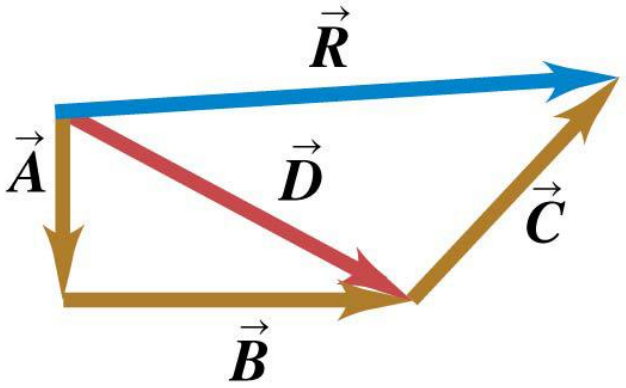
Adding More Than Two Vectors Graphically (1 of 3)

- To add several vectors, use the head-to-tail method.
- The vectors can be added in any order.

(a) To find the sum of these three vectors ...



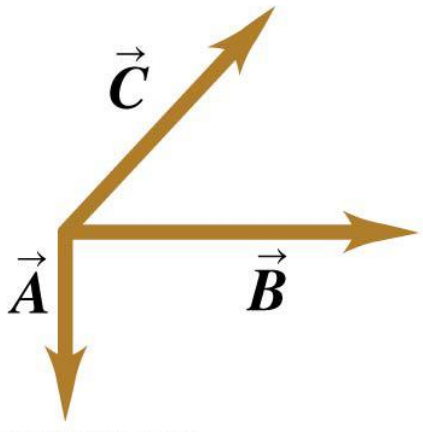
(b) ... add $\vec{A}$ and $\vec{B}$ to get $\vec{D}$ and then add $\vec{C}$ to $\vec{D}$ to get the final sum (resultant) $\vec{R}$ ...



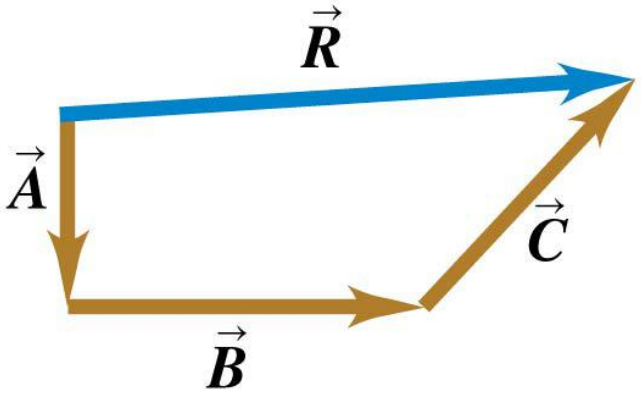
Adding More Than Two Vectors Graphically (2 of 3)

- To add several vectors, use the head-to-tail method.
- The vectors can be added in any order.

(a) To find the sum of these three vectors ...



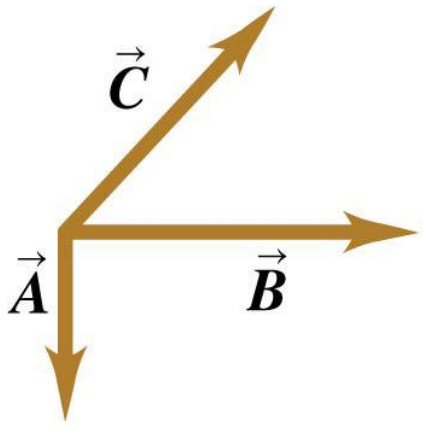
(d) ... or add $\vec{A}$, $\vec{B}$, and $\vec{C}$ to get $\vec{R}$ directly ...



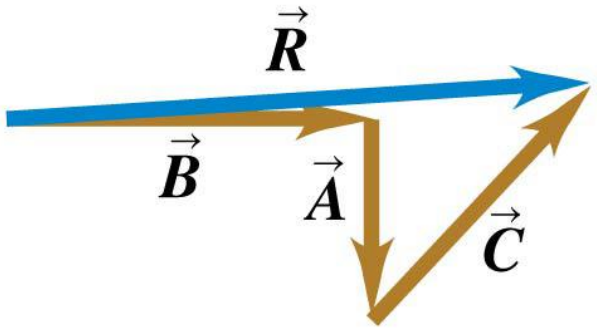
Adding More Than Two Vectors Graphically (3 of 3)

- To add several vectors, use the head-to-tail method.
- The vectors can be added in any order.

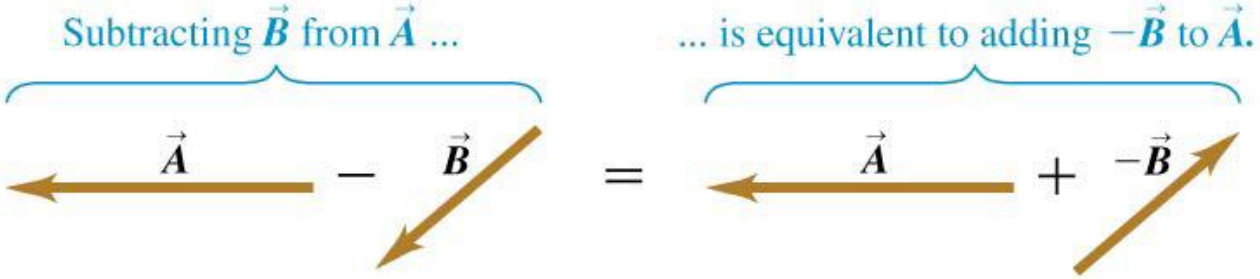
(a) To find the sum of these three vectors ...

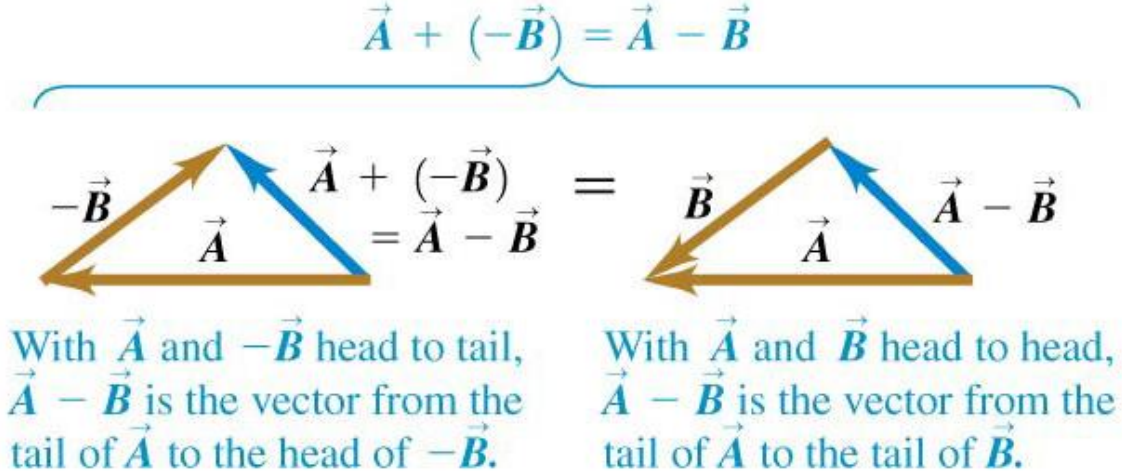


(e) ... or add $\vec{A}$, $\vec{B}$, and $\vec{C}$ in any other order and still get $\vec{R}$.



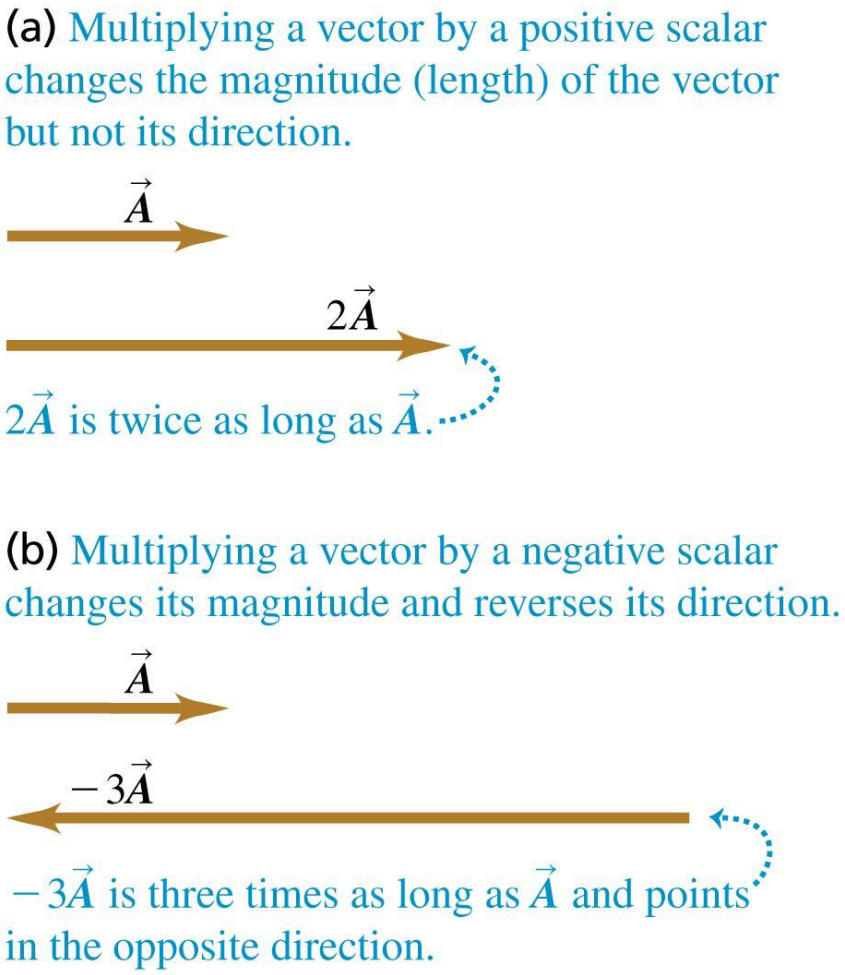
Vectors Subtraction





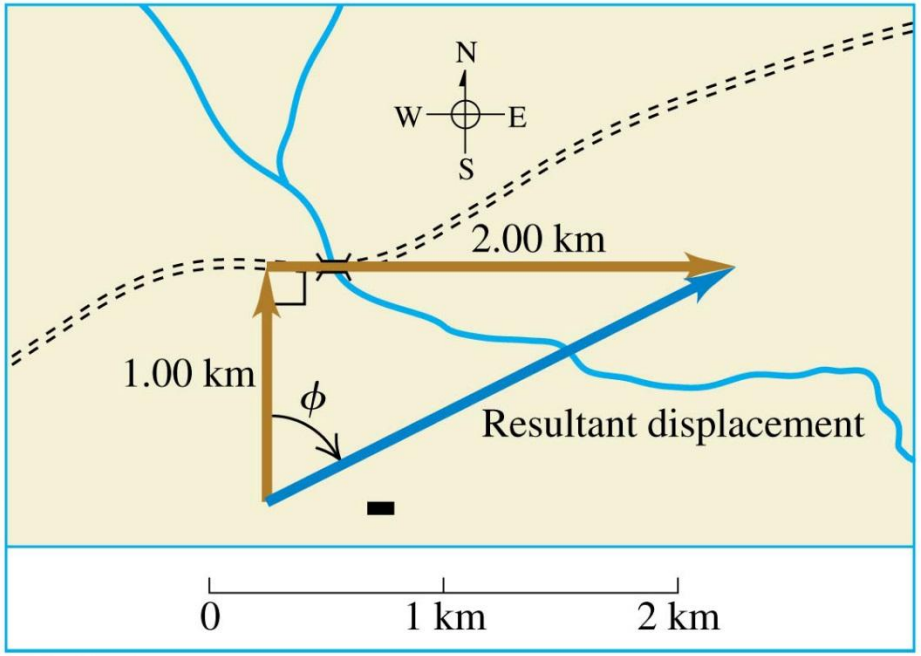
Multiplying a Vector by a Scalar

- If c is a scalar, the product $c\vec{A}$ has magnitude $|c|A$.
- The figure illustrates multiplication of a vector by
  a) a positive scalar and
  b) a negative scalar.



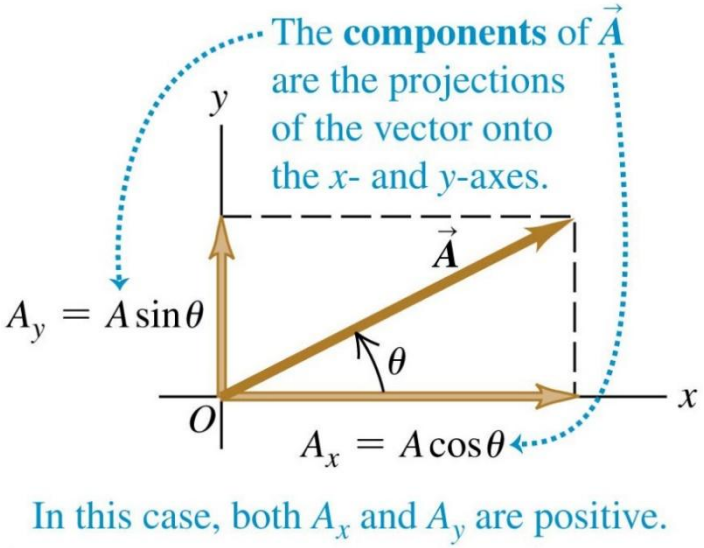
Addition of Two Vectors at Right Angles

- To add two vectors that are at right angles, first add the vectors graphically.
- Then use trigonometry to find the magnitude and direction of the sum.
- In the figure, a cross-country skier ends up 2.24 km from her starting point, in a direction of 63.4 degrees east of north.



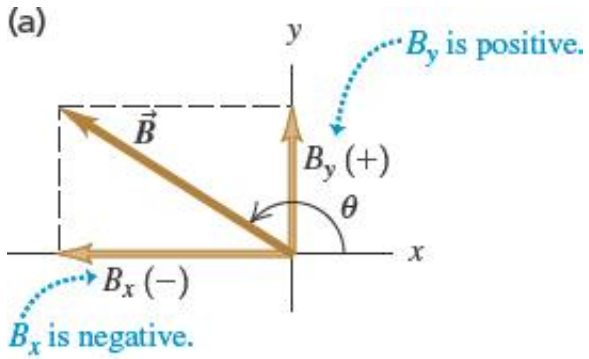
Components of a Vector

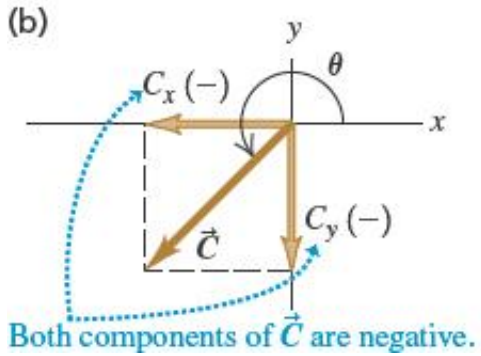
- Adding vectors graphically provides limited accuracy. Vector components provide a general method for adding vectors.
- Any vector can be represented by an *x*-component A_x and a *y*-component A_y.



Positive and Negative Components

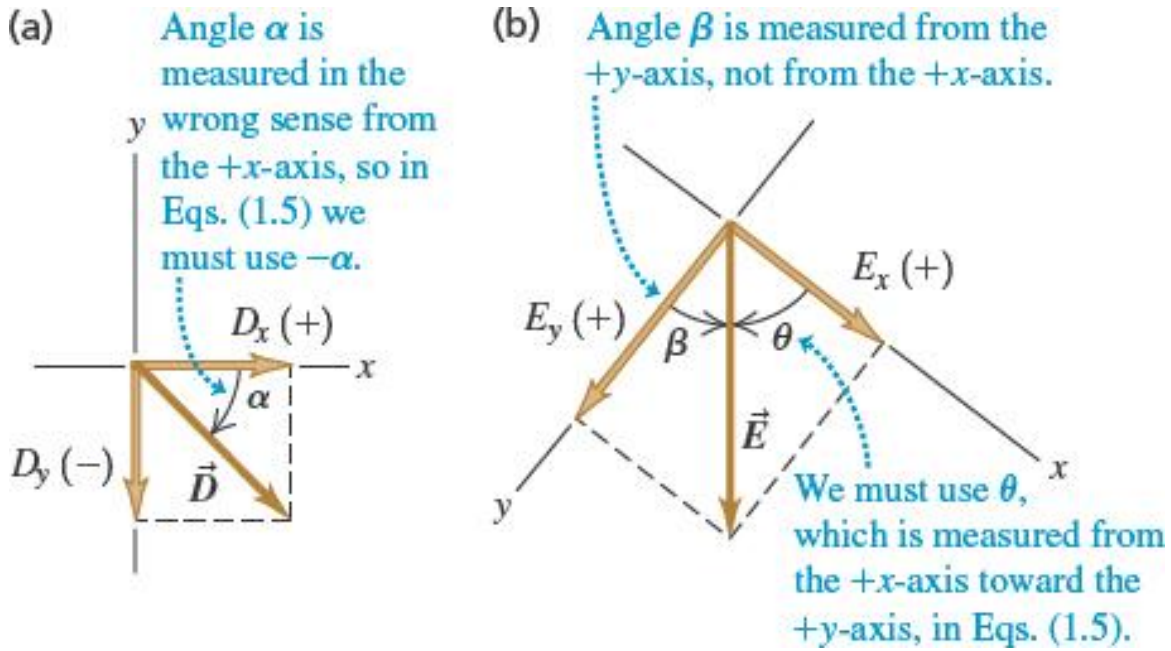
- The components of a vector may be positive or negative numbers, as shown in the figures.





Finding Components

- We can calculate the components of a vector from its magnitude and direction.

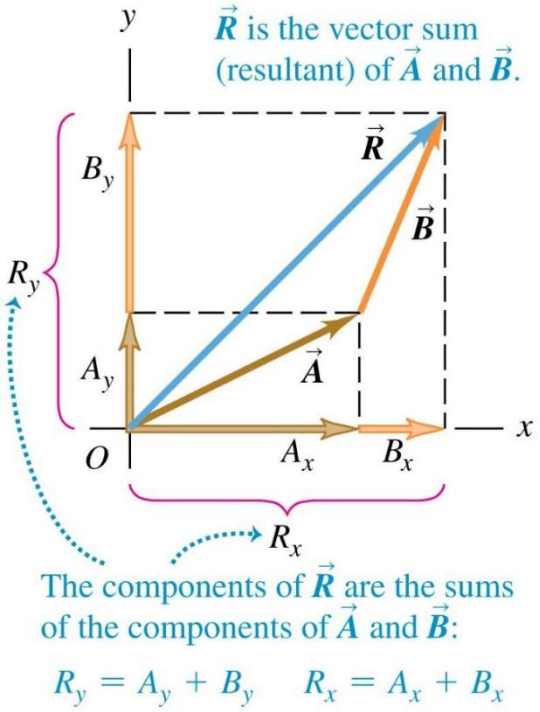


Calculations Using Components

- We can use the components of a vector to find its magnitude and direction: $A = \sqrt{A_x^{\,2} + A_y^{\,2}}$ and $\tan\theta = \dfrac{A_y}{A_x}$
- We can use the components of a set of vectors to find the components of their sum:

$$R_x = A_x + B_x + C_x + \ldots,\ R_y = A_y + B_y + C_y + \ldots$$

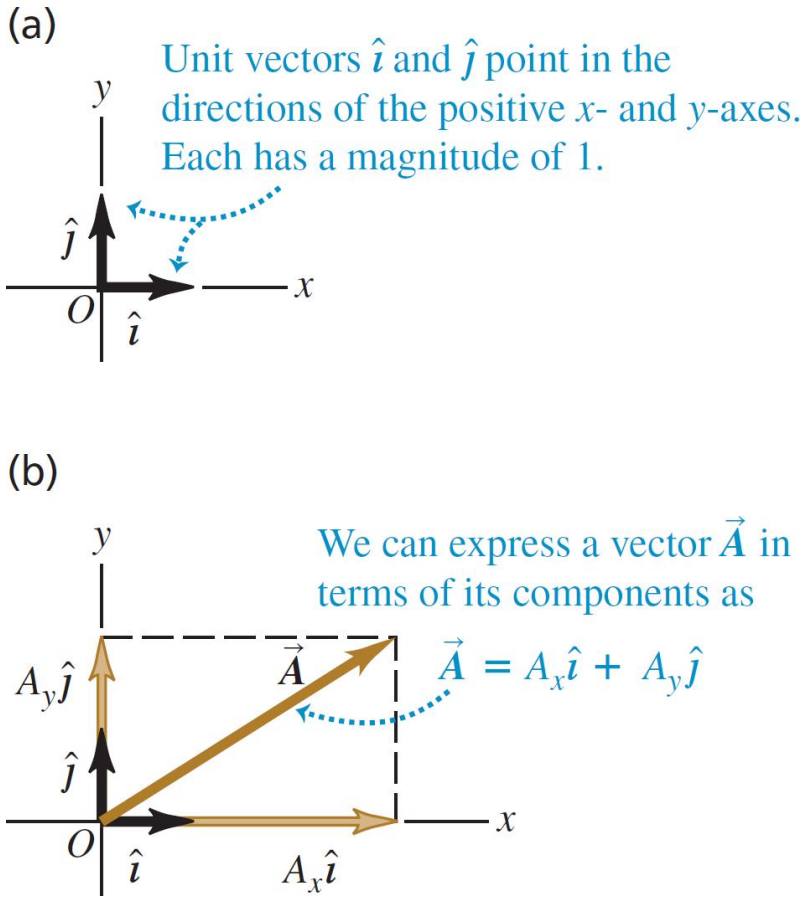
- Refer to **Problem-Solving Strategy 1.3**.



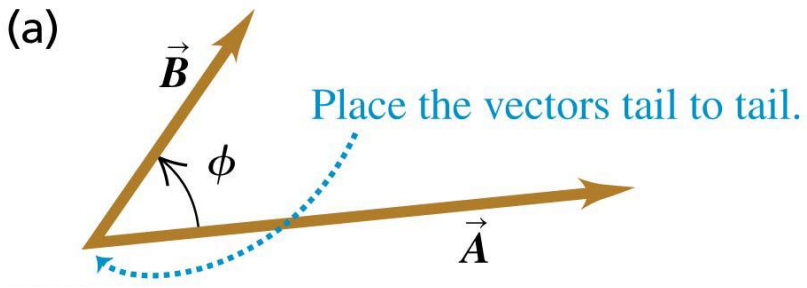
Unit Vectors

- A **unit vector** has a magnitude of 1 with no units.
- The unit vector $\hat{i}$ points in the +*x*-direction, $\hat{j}$ points in the +*y*-direction, and $\hat{k}$ points in the +*z*-direction.
- Any vector can be expressed in terms of its components as

$$\vec{A} = A_x\vec{i} + A_y\vec{j} + A_z\vec{k}$$



The Scalar Product (1 of 2)

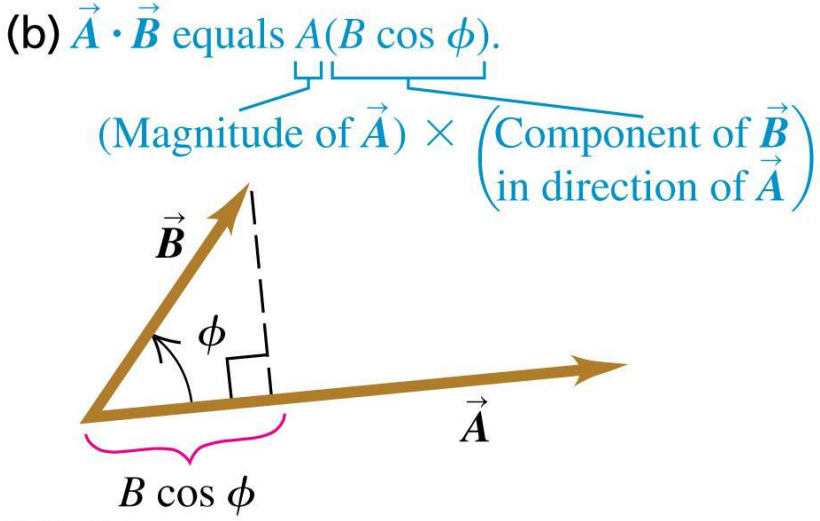


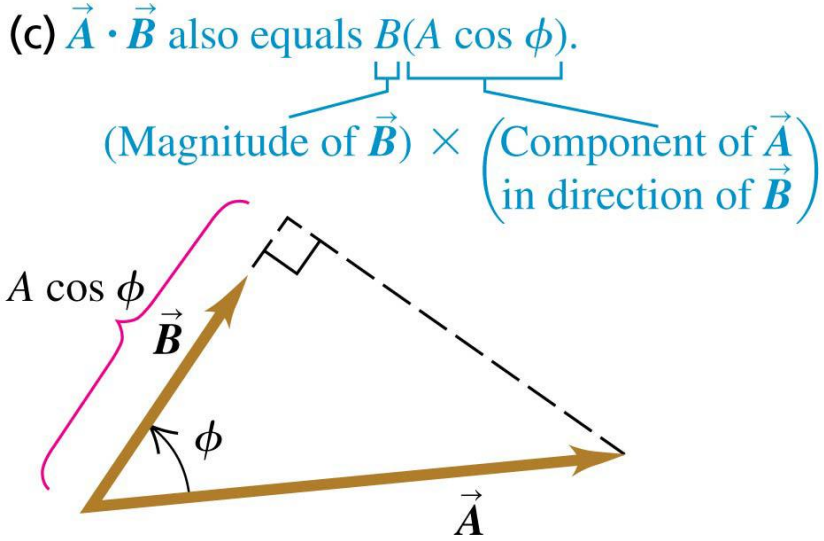
Scalar (dot) product of vectors $\vec{A}$ and $\vec{B}$

Magnitudes of $\vec{A}$ and $\vec{B}$

$$\vec{A} \cdot \vec{B} = AB\cos\phi = |\vec{A}||\vec{B}|\cos\phi$$

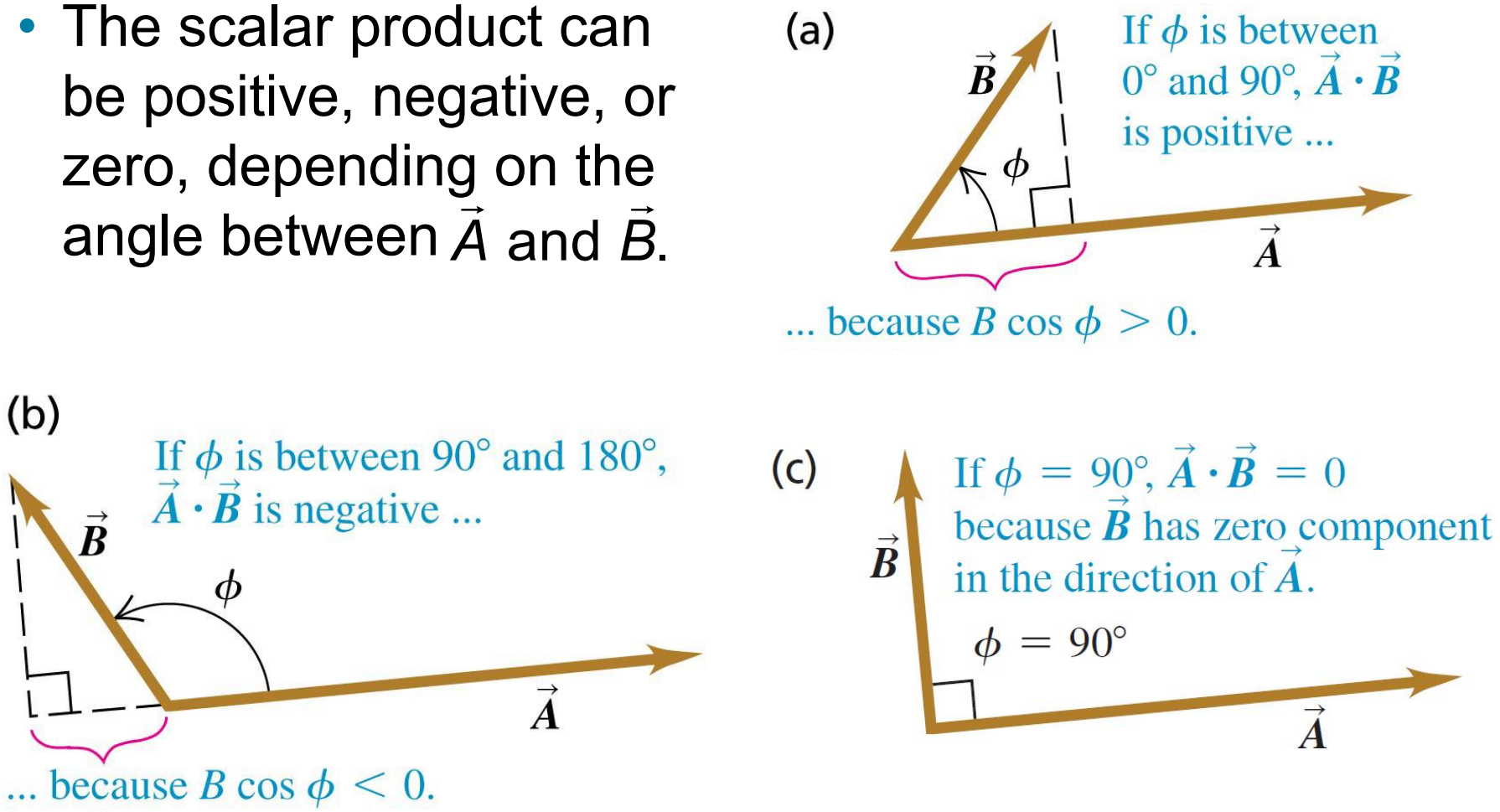
Angle between $\vec{A}$ and $\vec{B}$ when placed tail to tail





The Scalar Product (2 of 2)

- The scalar product can be positive, negative, or zero, depending on the angle between $\vec{A}$ and $\vec{B}$.



Calculating a Scalar Product Using Components

- In terms of components:

Scalar (dot) product
of vectors $\vec{A}$ and $\vec{B}$

Components of $\vec{A}$

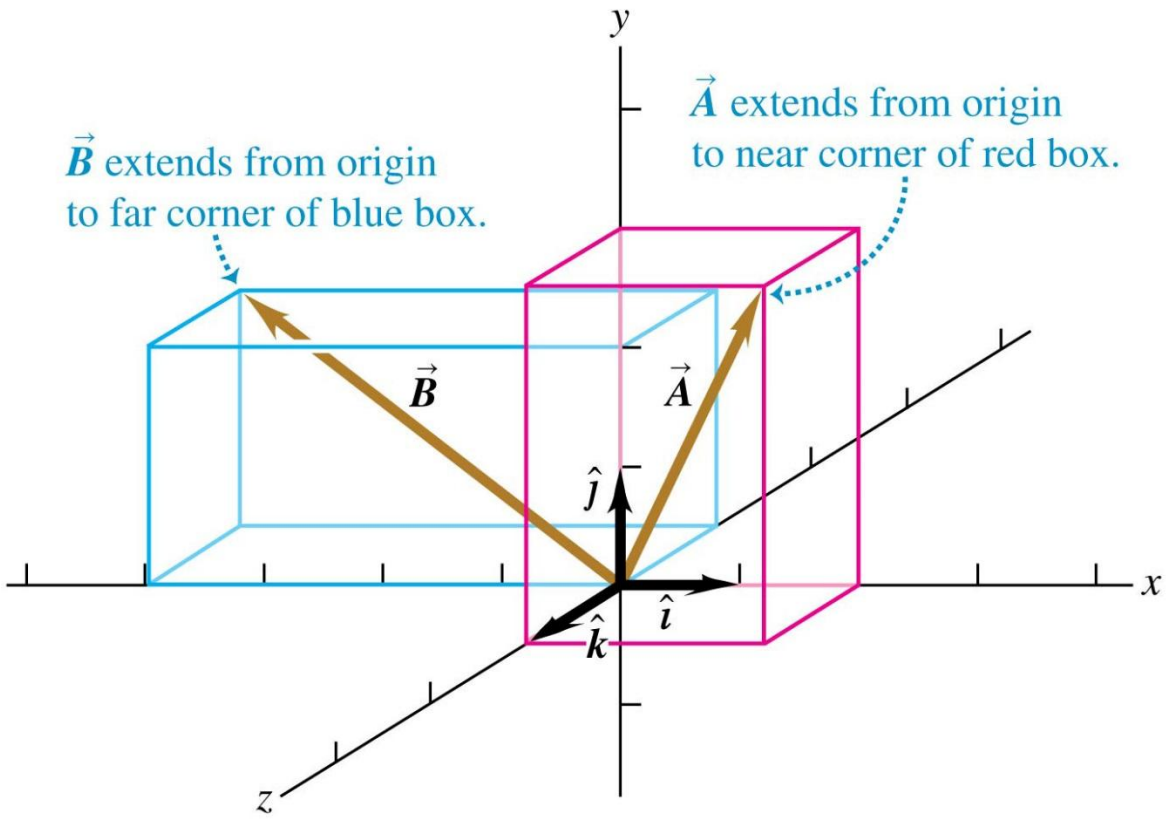
$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

Components of $\vec{B}$

- The scalar product of two vectors is the sum of the products of their respective components.

Finding an Angle Using the Scalar Product

- Example 1.10 shows how to use components to find the angle between two vectors.



Vector Product

If the vector product ("cross product") of two vectors is $\vec{C} = \vec{A} \times \vec{B}$ then:

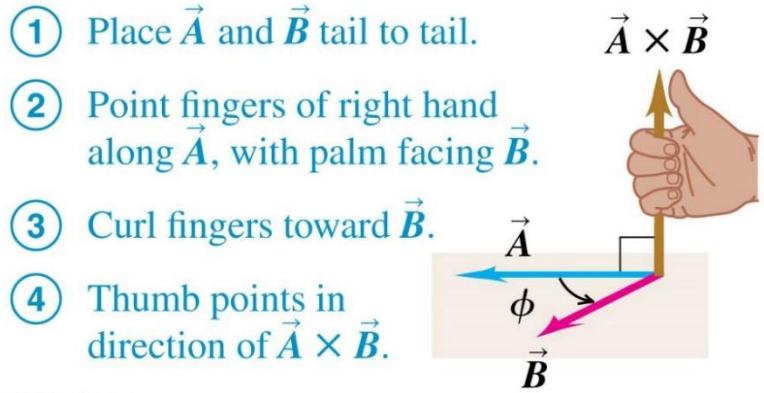
Magnitude of **vector (cross) product** of vectors $\vec{B}$ and $\vec{A}$

$$C = AB \sin\phi$$

Magnitudes of $\vec{A}$ and $\vec{B}$

Angle between $\vec{A}$ and $\vec{B}$ when placed tail to tail

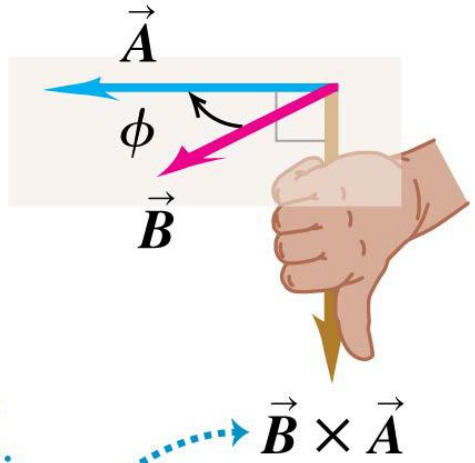
The direction of the vector product can be found using the right-hand rule:



The Vector Product is Anticommutative

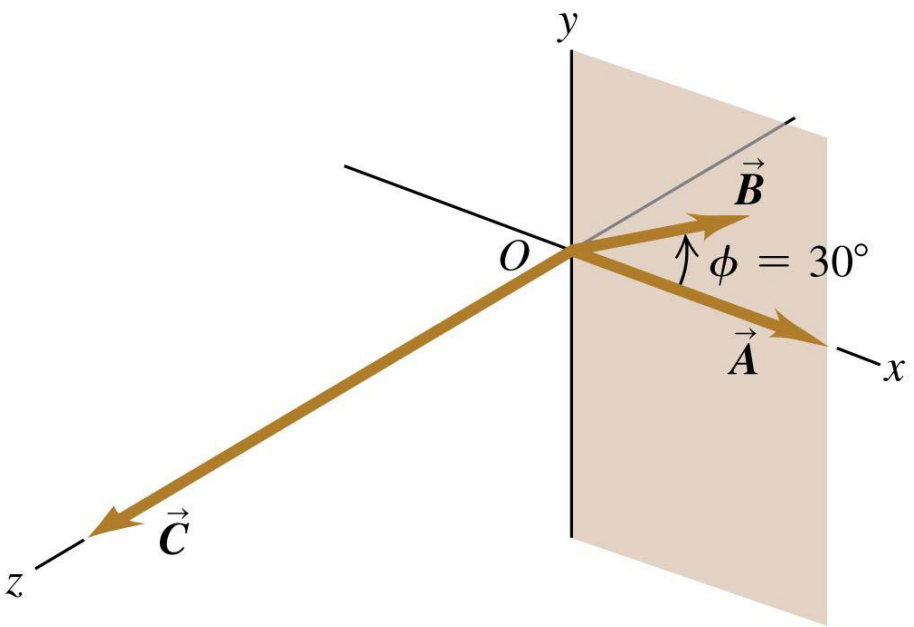
$$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$$

1. Place $\vec{B}$ and $\vec{A}$ tail to tail.
2. Point fingers of right hand along $\vec{B}$, with palm facing $\vec{A}$.
3. Curl fingers toward $\vec{A}$.
4. Thumb points in direction of $\vec{B} \times \vec{A}$.
5. $\vec{B} \times \vec{A}$ has same magnitude as $\vec{A} \times \vec{B}$ but points in opposite direction.



Calculating the Vector Product

- Use $AB\sin\phi$ to find the magnitude and the right-hand rule to find the direction.



1.7 Unit Vectors and Component Expression

$$\vec{A} = A_x \vec{i} + A_y \vec{j} + A_z \vec{k}$$

$$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$$

$$\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$$

$$\hat{i} \times \hat{j} = -\hat{j} \times \hat{i} = \hat{k}$$

$$\hat{j} \times \hat{k} = -\hat{k} \times \hat{j} = \hat{i}$$

$$\hat{k} \times \hat{i} = -\hat{i} \times \hat{k} = \hat{j}$$

Unit Vectors and Component Expression

$$\vec{A} \pm \vec{B} = (A_x \pm B_x)\ \hat{i} + (A_y \pm B_y)\ \hat{j} + (A_z \pm B_z)\ \hat{k}$$

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

$$A = \sqrt{\vec{A} \cdot \vec{A}} = \sqrt{A_x^{\ 2} + A_y^{\ 2} + A_z^{\ 2}}$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$= (A_y B_z - B_y A_z)\vec{i} + (A_z B_x - B_z A_x)\vec{j} + (A_x B_y - B_x A_y)\vec{k}$$

Unit Vectors and Component Expression

$$\frac{d\vec{A}}{dt} = (A_x\hat{i} + A_y\hat{j} + A_z\hat{k})'$$

$$= (A_x\hat{i})' + (A_y\hat{j})' + (A_z\hat{k})' = \frac{dA_x}{dt}\hat{i} + \frac{dA_y}{dt}\hat{j} + \frac{dA_z}{dt}\hat{k}$$

$$\int_0^t \vec{A}dt = \int_0^t (A_x\hat{i} + A_y\hat{j} + A_z\hat{k})dt$$

$$= \int_0^t (A_x\hat{i})dt + \int_0^t (A_y\hat{j})dt + \int_0^t (A_z\hat{k})dt = \left[\int_0^t A_x dt\right]\hat{i} + \left[\int_0^t A_y dt\right]\hat{j} + \left[\int_0^t A_z dt\right]\hat{k}$$